PARENT MODULE

School Management System

Technical Specification & Implementation Guide — v1.0 · March 2026

Confidential · Internal Documentation

Audience
Development Team · Product Team · School Administrators

Related Modules
Student Management · Academics · Finance · Communication · Attendance · Security

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1. MODULE OVERVIEW & SUBMODULE STRUCTURE

The **Parent Module** provides guardians with controlled access to their child's academic progress, financial obligations, and school communication.

The module is designed around the principle of:

Student autonomy + Parent transparency

Students receive **full detail access to their learning environment**, while parents receive **summaries and actionable information** without micromanaging student activities.

Parent Module Submodules

Submodule	Description	Priority	Status
Parent Authentication	Login and identity verification	Core	Active
Parent Dashboard	Overview of children performance and updates	Core	Active
Academic Summary	Performance summaries and exam results	Core	Active
Attendance Overview	Attendance trends and alerts	Core	Active
Finance & Payments	School fees, invoices and payment tracking	Core	Active
Communication	Messages, announcements and event alerts	Core	Active
Student Life	Clubs, sports and extracurricular activities	Enhanced	Active
Security & Access Logs	Parent account monitoring and audit	Core	Enhancement

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2. LOGIN & AUTHENTICATION

Purpose

Provide **secure role-based access** for both parents and students while ensuring guardians can only access their **linked children's records**.

Authentication Flow

```
Login Page
    │
    ├ Student Login
    │
    └ Parent Login
```

After authentication the system performs **role-based redirection**.

Role	Redirect
Student	Student Dashboard
Parent	Parent Dashboard
Staff	Staff Portal

Parent Login Credentials

Field	Value
Username	Registered phone or email
Password	Temporary password sent via SMS/email

Security rule:

Parent must **change password on first login**.

Student Login Credentials

Field	Value
Username	Admission number
Example	STU-2024-001
Password	Temporary password issued by school

Students may later update login details to:

- personal phone
- personal email

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3. EXTRA SECURITY PROTECTION

The system must enforce additional protection rules to safeguard student information.

Multi-Child Access

One parent account may link to multiple students.

Example:

```
Parent Account
│
├ Student A
├ Student B
└ Student C
```

Parents only see **students explicitly linked to their account**.

Divorced / Separated Parents

Each guardian receives a **separate account**.

```
Student
│
├ Parent Account 1
└ Parent Account 2
```

Both accounts access **the same student record**, but activity logs remain separate.

Session Security

Security controls include:

Feature	Rule
Session Timeout	30 minutes inactivity
Login Attempts	Limit to 5 attempts
Password Reset	Sent only to registered email/phone
Device Tracking	Record device/browser info

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4. ROLE-BASED ACCESS MODEL

The system separates **student view** and **parent view** to maintain balance.

Student Module (Direct Student Access)

Students access full learning information.

Features

Academic Information

- grades
- assignments
- exam schedules
- timetables

Learning Resources

- e-learning materials
- library access
- digital textbooks
- course resources

Student Activities

- clubs
- sports
- extracurricular events

Personal Progress

- attendance records
- academic performance analytics

Financial Information (Limited)

Students only see:

```
Fee Status
Paid / Pending
```

Students cannot see:

- invoice breakdown
- payment history
- family financial details

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5. PARENT DASHBOARD

Purpose

Provide guardians with **clear and summarized visibility** into their child's progress.

The dashboard acts as a **monitoring overview rather than a detailed student interface**.

Dashboard Widgets

Parent dashboard should include:

```
Child Overview
│
├ Academic Performance
├ Attendance Summary
├ Fee Balance
├ Upcoming Events
├ School Announcements
└ Teacher Messages
```

If parent has multiple children:

```
Child Selector
```

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6. ACADEMIC SUMMARY

Parents receive **high-level academic insights**.

Visible Information

Feature
Exam results
Term report cards
Performance trends
Subject averages

Example:

```
Math: 78%
English: 82%
Science: 74%
```

Parents can also see:

- improvement trends
- teacher remarks

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7. ATTENDANCE OVERVIEW

Parents view **attendance summaries instead of daily details**.

Information Displayed

Data
Monthly attendance percentage
Absence alerts
Late arrival patterns

Example:

```
Attendance Rate
92%

Absences
3 this month
```

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8. FINANCE & PAYMENTS

Parents manage all financial responsibilities.

Features

Financial Overview

Feature
Invoice list
Outstanding balances
Payment history
Receipts

Supported Payment Methods

The system should support African payment channels including:

- M-Pesa
- Airtel Money
- MTN MoMo

International payment options may include:

- Stripe
- PayPal

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9. COMMUNICATION

The Parent Module integrates with the **Communication System**.

Communication Channels

Parents receive:

- school announcements
- teacher messages
- meeting invitations
- event notifications

Notifications may be delivered via:

```
In-app notification
Email
SMS
```

Parents may also send:

- teacher inquiries
- meeting requests

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10. STUDENT LIFE OVERVIEW

Parents can see student participation in school life.

Information Displayed

Feature
Clubs joined
Sports teams
School competitions
Event participation

This allows guardians to monitor **holistic student development**.

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11. INFORMATION RESTRICTIONS FOR PARENTS

Certain information should remain **private to students or staff**.

Restricted Data

Parents cannot see:

Restricted Information
Detailed daily assignments
Private teacher notes
Minor disciplinary records
Internal classroom feedback
Library minor fines

Only **major disciplinary incidents** are escalated to parents.

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12. PERMISSION MATRIX

Role	Scope	Visibility
Student	Own record	Full detail
Parent	Linked children	Summaries
Staff	School records	Full detail
Admin	All data	Full access

Students cannot see:

```
siblings records
other students
```

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13. ACCESS TOKEN STRUCTURE (JWT)

Authentication uses **JSON Web Tokens**.

Example token:

```
{
 "sub": "user_id_123",
 "role": "parent",
 "student_ids": ["STU-2024-001","STU-2024-002"],
 "permissions": ["view_summary","view_finance"],
 "iat": 1710600000,
 "exp": 1710601800
}
```

Fields:

Field	Description
role	Determines dashboard redirect
student_ids	Limits data access
permissions	RBAC enforcement

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14. API ENDPOINTS

Parent Module endpoints should be mounted under:

```
/api/parent/
```

Parent APIs

Method	Endpoint	Description
GET	/api/parent/children/	List linked children
GET	/api/parent/dashboard/	Parent dashboard data
GET	/api/parent/academics/{student_id}/	Academic summary
GET	/api/parent/attendance/{student_id}/	Attendance summary
GET	/api/parent/finance/{student_id}/	Fee status
GET	/api/parent/payments/history/	Payment records

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15. SECURITY & ACCESS LOGGING

All parent actions must be logged.

Logged Events

Event
login_attempt
login_success
dashboard_view
student_record_access
payment_attempt

Log fields:

Field	Type
user_id	FK
action	String
ip_address	String
device	String
timestamp	Datetime

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16. INTEGRATION POINTS

Parent Module	Connected Module	Purpose
Parent	Student	Child records
Parent	Academics	Performance data
Parent	Attendance	Attendance summaries
Parent	Finance	Fees and payments
Parent	Communication	Messaging
Parent	Events	School calendar

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17. IMPLEMENTATION WORKFLOW

Account creation workflow:

```
Admission
     │
     ├ Student Account Created
     │
     └ Parent Account Created
            │
            ├ Parent receives login via SMS/email
            └ Student receives ID and password
```

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18. DESIGN PRINCIPLE

The Parent Module follows three core principles:

```
Student Autonomy
Parent Awareness
Platform Security
```

This ensures:

- students manage their learning
- parents remain informed
- schools maintain privacy boundaries